

Akash Kumar

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Skills

Kotlin, Java, Python, C++, Rust, SQL, JavaScript, Go, Android Native, Jetpack Compose, Flutter, HTML, CSS, AstroJS, Next.js, React.js, Supabase, WebSockets, Cloudflare.

Projects

FinSigma — Full-Stack Simulated Trading Platform

finsigma.vercel.app

Next.js, TypeScript, PostgreSQL, Prisma, Tailwind CSS, JWT

- Engineered an ACID-compliant simulated trading platform with live market pricing to strictly enforce balances and prevent ledger desynchronization during high-concurrency trades.
- Optimized financial instrument discovery by implementing a millisecond-latency fuzzy-search engine using PostgreSQL trigram similarity operators (<% and <<->).

CloudClock — Cloud-Synchronized Alarms

github.com/skyprincegamer/cloud_clock

Kotlin, Jetpack Compose, Supabase Realtime, Room DB, Android AlarmManager

- Eliminated alarm polling and achieved instantaneous bidirectional sync across 3+ concurrent physical and emulated devices using a cloud-sync engine triggered by Supabase Realtime.
- Guaranteed Doze-mode-resilient alarm delivery using localized UTC conversions and precise RTC_WAKEUP intents.

Oxy Bars — GPU Accelerated Audio Spectrum Analyzer

github.com/skyprincegamer/oxy_bars

Rust, cpal, rustfft, Macroquad, TOML

- Solved spectral leakage and ambient noise pollution in real-time audio analysis by implementing Blackman-Nuttall windowing and live spectrum subtraction.
- Processed 44.1kHz audio streams with sub-16ms visual render cycles, achieving 22µs lock-free FFT pipeline cadences via atomic memory ordering.

Open Source Experience

KDE KleverNotes — QML/C++ Note-Taking App

[KDE Invent](#)

- Engineered a dynamic note-sorting engine with preemptive state-saving to prevent data loss during live UI configuration changes (MR !171).
- Implemented a draggable canvas pan tool with coordinate bound limits to prevent out-of-bounds rendering on the painting page (MR !186).
- Expanded painting capabilities by developing robust shape-drawing features and real-time elliptical previews (MR !176).
- Resolved critical file path resolution bugs for saved painted images, contributing 240+ lines of robust QML and C++ across all 4 MRs (MR !175).

Education

Integrated B.Tech + M.Tech in Information Technology

Expected Graduation: May 2029

ABV-IIITM Gwalior (*Atal Bihari Vajpayee Indian Institute of Information Technology and Management*)

CGPA: 8.18

Class XII (ISC) — 95.74%

Class X (ICSE) — 97.80%